

Short Communication

Symbiodiniaceae microRNAs and their targeting sites in coral holobionts: A transcriptomics-based exploration

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ARTICLE INFO

Keywords:

Reef-building coral
Symbiodiniaceae
microRNA
Transcriptome

ABSTRACT

Corals should make excellent models for cross-kingdom research because of their natural animal-photobiont holobiont composition, yet a lack of studies and experimental data restricts their use. Here we integrate new full-length transcriptomes and small RNAs of four common reef-building corals with the published *Cladocypium* genomes to gain deeper insight into gene regulation in coral-Symbiodiniaceae holobionts. Eleven novel Symbiodiniaceae miRNAs get identified, and enrichment results of their target genes show that they might play a role in downregulating rejection from host coral cells, protecting symbiont from autophagy and apoptosis in parallel. This work provides evidence for the early origin of cross-kingdom regulation as a mechanism of self-defense autotrophs can use against heterotrophs, sheds more light on coral-Symbiodiniaceae holobionts, and contributes valuable data for further coral research.

1. Introduction

miRNA is a type of small, single-stranded non-coding RNA found in plants, animals, and some viruses, and that functions in RNA silencing and post-transcriptional regulation of gene expression [1]. miRNA is recognized to be involved in various regulation pathways, such as cross-cell regulation, cross-organ regulation, and even cross-generation regulation, however, whether cross-kingdom regulation occurs has been controversial. In 2012, Zhang and colleagues first proposed this concept after they identified stable plant miRNAs in several human and animal organs [2]. They argued that intaken plant miRNAs could remain active and exert biological effects in animal tissues. A great deal of subsequent research has supported their views, for example, [3] revealed a correlation between fruit juice intake and fruit miRNA concentration in human serum; one honeysuckle miRNA [4] and two rapeseed miRNAs [5] were observed to be effectively absorbed by mice; [6] integrated ddPCR and transcriptome analyses to explore silkworms' absorption of mulberry miRNAs; etc. Some "avid" supporters of the idea of cross-kingdom miRNA activity advanced the medicinal value of plant miRNA against IAV [4], tumors [7], and even SARS-CoV-2 [8]. Nevertheless, ever since the concept was put forward, the doubts continue to be voiced [9,10]. The proposers and supporters of cross-kingdom

regulation have been working on explaining miRNA absorption mechanisms in detail in order to refute these objections [11], while not all research of interorganismal miRNA exchange is focused on food-intake-derived regulation, such as [12] studied the role of dodder miRNA in the parasitic process.

Most studies on cross-kingdom miRNA activity study humans or other complex animals as experimental subjects and look into whether plant miRNAs can affect animals' physiological functions. From our perspective, these models may be too complex in terms of behavior and physiology to support accurate conclusions about the origins or even the existence of such small molecules. Instead, we suggest a focus on more basal species exhibiting less complex behavior to explore the origins and details of the dominant regulation mechanisms of miRNA, which could lay the theoretical and practical foundation for studies of more complex organisms. We recommend coral as a potential animal model for cross-kingdom regulation research. Corals obtain the majority of their energy and nutrients from photosynthetic unicellular dinoflagellates, commonly known as symbiotic Symbiodiniaceae [13]. Symbiodiniaceae living within corals' tissues not only supply their hosts with the products of photosynthesis but also aid in coral calcification and waste removal. Coral-Symbiodiniaceae holobionts are naturally formed "animal-plant mixtures", that offer the advantages of simple behavior and ease of

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<https://doi.org/10.1016/j.ygeno.2022.110404>

Received 17 March 2022; Received in revised form 30 May 2022; Accepted 9 June 2022

Available online 14 June 2022

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observation relative to more complex animal or plant models. It is not difficult to assume that cross-kingdom regulation, regarded by its proposers as evidence of long-term co-evolution between photosynthetic organisms and animals, may play an important role in these holobionts. In fact, inter-partner signaling in cnidarian-dinoflagellate symbiosis has been mentioned decades ago [14], and there have been successful attempts to explore cross-kingdom regulation in *Aiptasia* models [15,37] analogous to coral symbiosis.

However, in contrast to more common model organisms, there is a lack of public omics data from reef-building corals, not commensurate with their important status in the marine ecosystem and scientists' sharp focus on them. At the end of 2020, only 24 coral genomes at a medium assembly level are available on NCBI, while transcriptomes and proteomes are even more lacking. Many other well-known databases, such as miRBase [16] and KEGG [17], contain very little or even no coral data. Various factors, including geographic location, technical limitations, and experimental challenges, contribute to the limited number of coral studies serving as data sources, resulting in a "data gap". In-depth research underpinned by high quality bio-data is in urgent demand. A few data portals that focus on marine data such as Reefgenomics [18] and OIST Marine Genomes (<http://marinegenomics.oist.jp>) have been developed, yet more efforts should be made to fill in this gap.

In order to explore intracellular regulation of coral-Symbiodiniaceae holobionts, we sequenced the full-length transcriptomes and small RNAs of four common and frequently dominant reef-building corals from the Xisha Islands in the South China Sea, including *Acropora muricata*, *Montipora foliosa*, *Montipora capricornis* and *Pocillopora verrucosa*. Every full-length transcript was annotated with Nr, Nt, Pfam, KOG, Swiss-Prot, GO and KEGG to distinguish coral and Symbiodiniaceae genes. According to Coral Trait [19] records and previous studies [20,21] that the majority of Symbiodiniaceae in investigated corals belong to *Cladocopium*, published *Cladocopium* genomes [22–24] were employed to identify Symbiodiniaceae miRNAs, whose targeting sites were predicted later. Finally, miRNA-target genes were selected for enrichment analysis. Results demonstrate that some of the Symbiodiniaceae miRNAs target immune-related genes in corals, suggesting that cross-kingdom regulation be a way for the Symbiodiniaceae to prevent rejection by their hosts. This work sheds light into the coral-Symbiodiniaceae holobiont, an ancient association in which an early example of cross-kingdom regulation is found. We also curate our data in a full-service online database in order to support further research (<http://coral.bmeonline.cn>).

2. Results

2.1. Gene functional annotation

The offline polymer read bases of *A. muricata*, *M. foliosa*, *M. capricornis* and *P. verrucosa* samples are 12.6G, 20.16G, 16G and 14.97G respectively, and down-stream processing details are available in File S1. Unigene numbers are 14,926 in *A. muricata*, 14,040 in *M. foliosa*, 11,410 in *M. capricornis* and 12,982 in *P. verrucosa*. Their sequences are available in Data S1.

The full-length transcriptomes were annotated with the aid of Nr, Nt, Pfam, KOG, Swiss-Prot, GO and KEGG databases, and related unigene statistics are shown in Fig. 1. In function-related databases, over 70% of unigenes get GO and KEGG annotations, paving the way for functional enrichment analysis. In protein-related databases, more than 96% of unigenes are annotated in Nr, the basic protein primary sequence database. As illustrated in Fig. S1.A, Nr annotations map most genes of the four corals to the cnidarians *A. digitifera*, *E. pallida* and *N. vectensis*, with more than 85% overlap. Acceptable as it is, this result still reflects the lack of coral data for those annotations rarely correspond with the actual species in question. All annotation details are available in Data S2.

According to Nr annotation, there are only about 400

Symbiodiniaceae genes in four investigated corals, less than 0.075% of the total gene number. Since Nr database may not cover newly published Symbiodiniaceae genes, transcripts were aligned to gene models of *Cladocopium C1* [22], *Cladocopium C15* [23], *Cladocopium C92* [24] and *Durudinium trenchii* [41] for correction and supplement. The number of re-identified Symbiodiniaceae gene is 58 in *A. muricata*, 212 in *M. foliosa*, 67 in *M. capricornis* and 178 in *P. verrucosa*, where 90 genes keep their initial Nr annotations (Fig. S1.B, Table S1). It seems that Symbiodiniaceae genes in four investigated corals have relatively high sequence similarity with those of *Cladocopium C1*, implying the potential homology between them.

2.2. Symbiont miRNA identification

The raw small RNA read bases are 4.412G (1.548G + 1.359G + 1.505G) from *A. muricata*, 3.865G (1.275G + 1.053G + 1.537G) from *M. foliosa*, 3.295G (1.177G + 0.893G + 1.225G) from *M. capricornis* and 6.235G (2.67G + 1.494G + 2.071G) from *P. verrucosa*, and down-stream processing details are available in File S2. Since there were few mature solutions for directly acquiring Symbiodiniaceae genomes from the investigated corals, published *Cladocopium* genomes, including *Cladocopium C1*, *Cladocopium C15* and *Cladocopium C92*, were employed to identify Symbiodiniaceae miRNAs by the aid of miRDeep-P2. Eleven Symbiodiniaceae miRNAs are identified from the total small RNA, as shown in Table 1, numbered as cgo-miR-tmp-1 to 11. These novel Symbiodiniaceae miRNAs do not overlap with any published Symbiodinium [25] or coral [26,27] miRNA, and they are unlikely to be of coral origin, since no sequences homologous to their locus are found on published genomes of *A. muricata* (GCA_014634545.1), *P. verrucosa* (GCA_014529365.1) or *Montipora* corals (GCA_014634245.1, GCA_014634505.1 and GCA_006542545.1).

Their expression profiles calculated by miRDeep2 are supplemented in Table S2. Four miRNAs, cgo-miR-tmp-1, 2, 4, and 7, exist in all investigated corals, and their expression levels are relatively high. It is as expected that, in intracellular environments, photobiont miRNA can maintain a certain concentration beyond those observed in traditional cross-kingdom research [2–8]. Associating Table S2 with Table 1, cgo-miR-tmp-1, 2, 5, 7 and 8 may be the conservative miRNAs existing in many *Cladocopium* species, while most miRNAs non-co-expressed among corals are only found in one *Cladocopium* genome (See File S3 for loci details).

2.3. Target analysis

Targeting sites of Symbiodiniaceae miRNAs were predicted using miRanda. Fig. 2 demonstrates the statistics of target gene number for each miRNA in investigated corals. The targeting range of cgo-miR-tmp-7, which has the highest expression level in all samples, presents to be extremely broad. It hits on up to 19.5% of total genes, and its target number far exceeds those of other miRNAs. The only exception occurs in *M. foliosa*, where this coral's specific miRNA cgo-miR-tmp-10 has the most target genes. It's also found that cgo-miR-tmp-7 shares many targets with others, the symbol of synergistic action. Details of all targeting sites are available in Data S3.

Target genes were all selected for GO/KEGG enrichment, results of which are listed in Table S3. 121 GO terms and 92 KEGG pathways get significantly enriched, covering quite a few functions. Several enrichment results of representative miRNA are picked out and highlighted in Fig. 3. *M. foliosa* has a unique miRNA group consisting of cgo-miR-tmp-3, 6, 8, 9 and 10. In the results of enrichment analysis of their target genes, a series of GO terms related to the cytoskeleton emerge, among which Rho-related terms are notable. Two terms, regulation of Rho protein signal transduction (GO:0035023) and Rho guanyl-nucleotide exchange factor activity (GO:0005089), are also enriched in the target genes of cgo-miR-tmp-1 in *M. capricornis* and cgo-miR-tmp-7 in *P. verrucosa* (Table S3). For *A. muricata*, there is one non-co-expressed

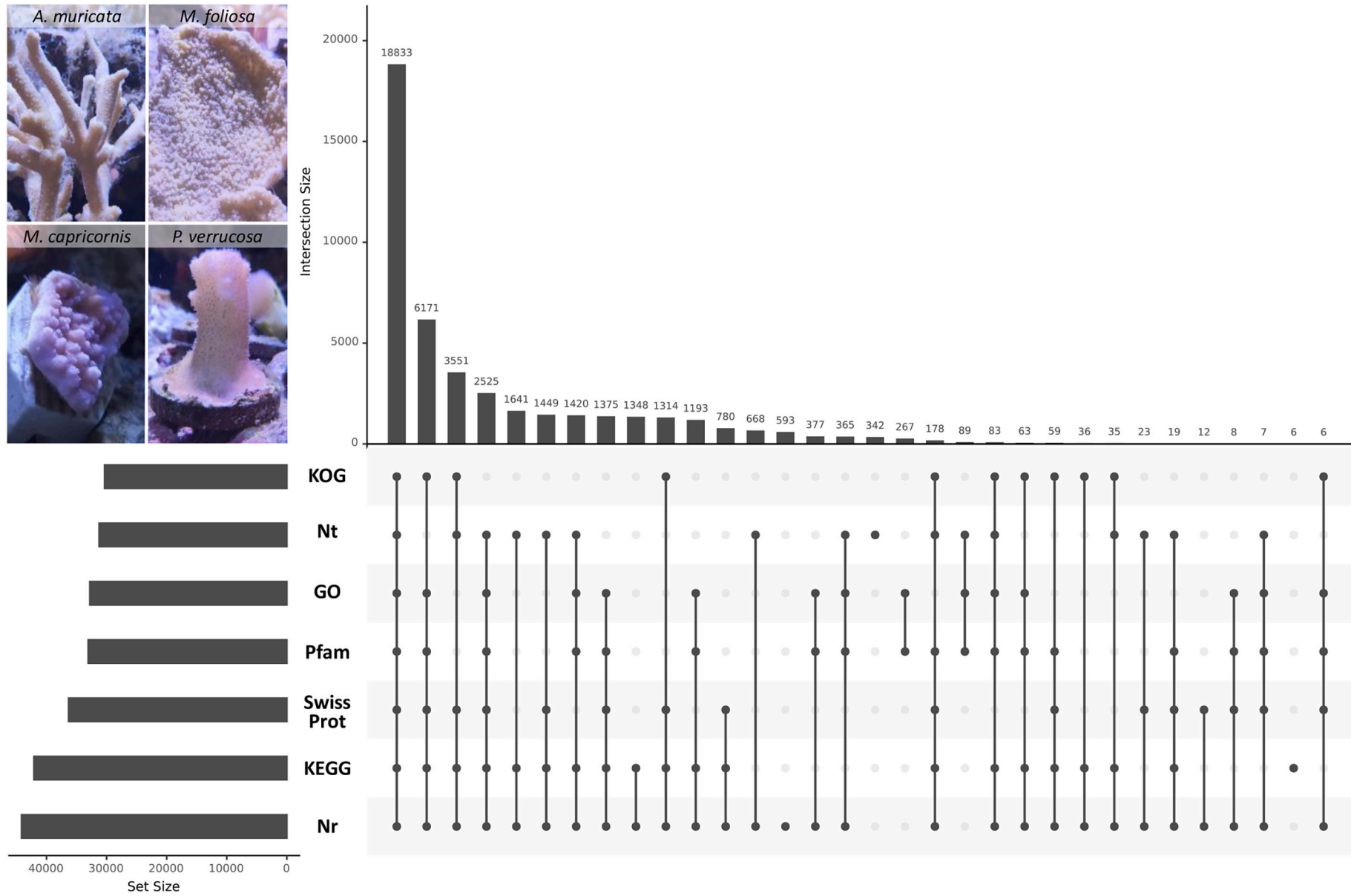


Fig. 1. Statistics of gene functional annotation in the form of Upset plot. Photos of investigated corals are listed in the upper left corner. Each column corresponds to a database, and bar charts on top show the annotation number from the database. Each row corresponds to a possible intersection: the filled-in cells show annotation from which database is part of an intersection.

Table 1

Basic information of eleven novel Symbiodiniaceae miRNAs. Details of these miRNAs, such as sequences, genomic loci, secondary structures of their precursors and targeting sites, are available in **Data S3** as well as online database CoralSym (<http://coral.bmeonline.cn/sym/mature/NOVEL>).

miRNA	Sequence	Loci on genomes		
		<i>Cladocopium C1</i>	<i>Cladocopium C15</i>	<i>Cladocopium C92</i>
cgo-miR-tmp-1	ACAAGCGUGGCUAGUUGACU	scaffold1282	NODE_143743	BGPT01000008.1
cgo-miR-tmp-2	ACAGAUUGCGGCAACCGUGCAG	scaffold532	scaffold_14538	BGPT01002003.1
cgo-miR-tmp-3	CAGGCUGUUGCUUUUGGAGUGA	scaffold684	/	/
cgo-miR-tmp-4	GCGGGUUGCUGCCAUCUGUU	scaffold301	scaffold_8115	/
cgo-miR-tmp-5	GGAUGUCGGCGAUGUGGAUG	scaffold3593	scaffold_1182	BGPT01000643.1
cgo-miR-tmp-6	GUAAAGCUCUGAGAAUUGGC	scaffold11300	/	/
cgo-miR-tmp-7	UGAAAUCUUUGCUCUGAAGUG	scaffold11203	NODE_117889	BGPT01003029.1
cgo-miR-tmp-8	UGGCCCCCUGCUGGCCAAUCUG	scaffold585	scaffold_1449	BGPT01001549.1
cgo-miR-tmp-9	UGGCUUACUGUUUGUUUUGUU	scaffold6920	/	/
cgo-miR-tmp-10	UUCAUCUGCCGCGUUGCUAGC	scaffold13821	/	/
cgo-miR-tmp-11	UUCGAAUCUUCUAGUUAGC	scaffold9207	/	/

Symbiodiniaceae miRNA, cgo-miR-tmp-11. Several cytoskeleton-related terms appear in GO enrichment results of its target genes, while KEGG enrichment results directly demonstrate pathways in immune system.

3. Discussion

3.1. Anti-autophagy

Rho GTPases belonging to the Ras superfamily of small GTP-binding protein serve as “molecular switches” in various signaling pathways, mainly by exerting effects on the cytoskeleton [28]. They are widely distributed in immune-related cells and are frequently observed to participate in immune regulation. They usually cycle between an active GTP-bound state and an inactive GDP-bound state, triggering immune responses when bound to GTP until they are hydrolyzed into the GDP-bound state. Three proteins are recognized to regulate this cycle, among which Rho guanyl-nucleotide exchange factor (Rho-GEF) contributes to the activation of Rho GTPases, playing the role of positive regulator. From this it is assumed that Symbiodiniaceae miRNAs might downregulate the activity of positive regulators in immune pathways. These miRNAs may also help remodel the cytoskeleton of corals to allow for the entry of Symbiodiniaceae, in a similar way to how some parasites affect their hosts [29,30].

In addition, autophagy-related genes are identified as these miRNAs' targets as well, which may strengthen the credibility of the assumptions outlined above in view of many observations that Rho GTPases upregulate autophagosome fusion and transport [31–34]. Fig. 4A displays a predicted pathway by which Symbiodiniaceae block autophagosome production via miRNA-silencing the upstream Rho “switch”, nevertheless the possibility that Symbiodiniaceae miRNAs directly target autophagy-related genes cannot yet be ruled out.

3.2. Anti-apoptosis

Beyond autophagy, host cell apoptosis also threatens the survival of Symbiodiniaceae [35–37]. The NF-κB signaling pathway highlighted in Fig. 3 deserves attention, for cnidarian host will mount an immune response to eliminate the stressed symbiont if NF-κB is upregulated. Two substances were recorded to regulate NF-κB, sphingosine 1-phosphate (S1P) formed by sphingosine kinase and sphingolipids sphingosine (Sph) formed by the action of sphingosine-1-phosphate phosphatase (SPPase) [38,39]. The former promotes cell survival and proliferation, while the latter pushes a cell toward apoptosis and death. Highly co-expressed cgo-miR-tmp-7 is found to be involved with these two. As listed in Table 2, cgo-miR-tmp-7 might be able to inhibit the expression of sphingosine-1-phosphate lyase (SPL) for S1P degradation [40] and SPPase for Sph production, protecting Symbiodiniaceae from host coral cell apoptosis. The graphical presentation of this predicted pathway is shown as Fig. 4B.

It should be mentioned that genes related to SPPase and SPL are counted as rare in coral holobionts. In *A. muricata*, there are three isoforms of SPPase XP_015778776.1, one of which is under the joint effect of cgo-miR-tmp-7 and cgo-miR-tmp-8; in *M. foliosa* and *M. capricornis*, all SPLs including the isoforms of KXJ12320.1 and XP_015767852.1 belong to the targets of cgo-miR-tmp-7; in *P. verrucosa*, cgo-miR-tmp-7 has targeting sites on the one and only SPPase (KXJ25723.1). In contrast, sphingosine kinases forming S1P in all investigated corals are not targeted by any Symbiodiniaceae miRNAs. These above might reveal their targeting selectivity in anti-apoptosis process to some extents.

3.3. Limitation

This work is typical omics-based research, mining information from large-scale bio-data, thus it is inevitably faced with systematic errors coming from bioinformatics tools and databases, especially when taking the “coral data gap” into account. However, the largest limitation to this research project actually comes from the lack of data about Symbiodiniaceae. Although there are indeed successful attempts to directly isolate Symbiodiniaceae genomes from several coral hosts, it lacks similar feasible solutions for our South China Sea reef-building corals, thus even using all existing Symbiodiniaceae genomes for miRNA identification couldn't avoid omissions. It has been observed that Clade D Symbiodiniaceae also occupy a certain percentage of the *P. verrucosa* holobiont [19,20], implying that a great deal of information is yet to be mined. One *Durudinium* genome has been already published [41], yet more data are in demand for accuracy.

4. Conclusion

In summary, our work integrates new full-length transcriptomes and small RNAs of four common and frequently dominant reef-building corals with public *Cladocopium* genomes to explore coral-Symbiodiniaceae holobionts, shedding light on the interaction between these ancient heterotrophs and autotrophs. It reveals the potential miRNA-based anti-rejection mechanism used by Symbiodiniaceae, and offers a foundation for further cross-kingdom research.

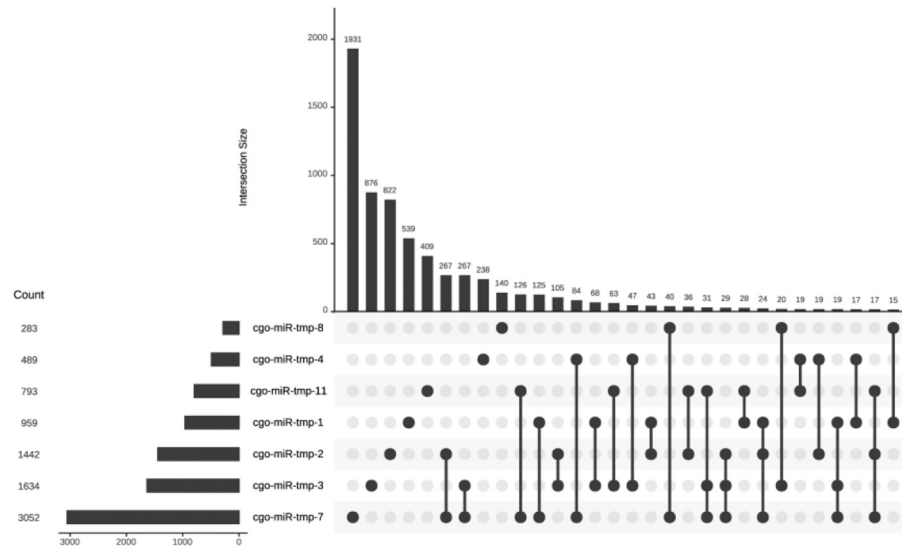
We have added all our data, including gene annotation and miRNA information, to the CoralBioinfo database (<http://coral.bmeonline.cn>), in a gesture to build a high-quality multi-omics platform for further coral research. Multiple web services have been developed for it, and data contribution requests for joint efforts are always welcome.

5. Materials and methods

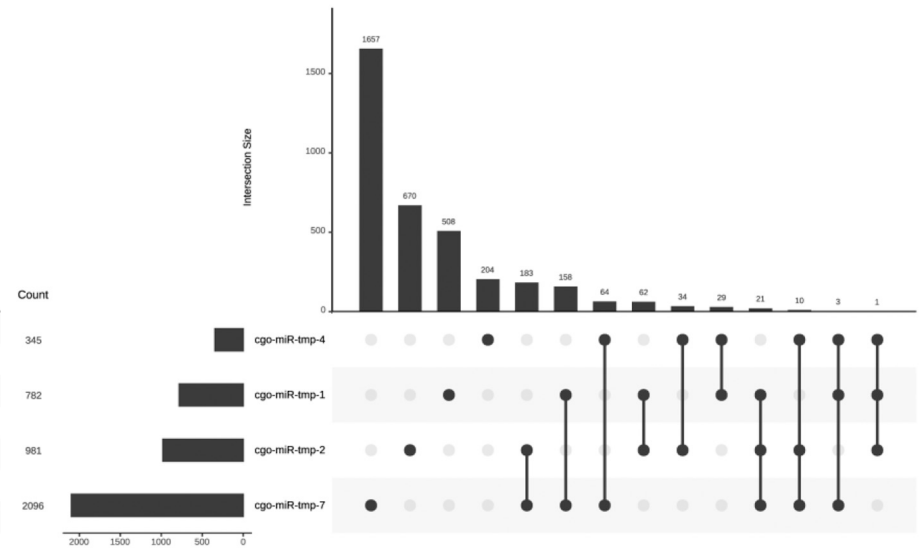
5.1. Experimental model and subject details

All coral samples were collected and processed in accordance with local laws for invertebrate protection and approved by the Ethics

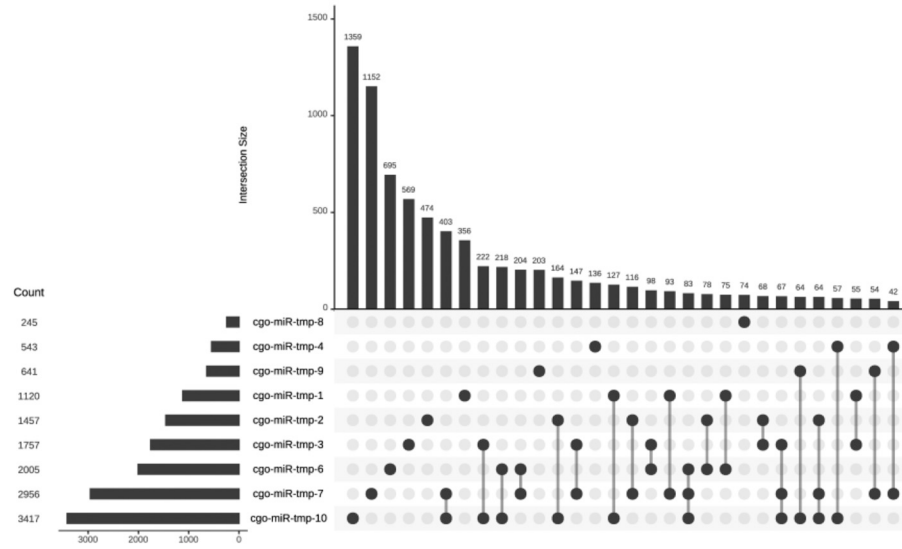
A. muricata



M. capricornis



M. foliosa



P. verrucosa

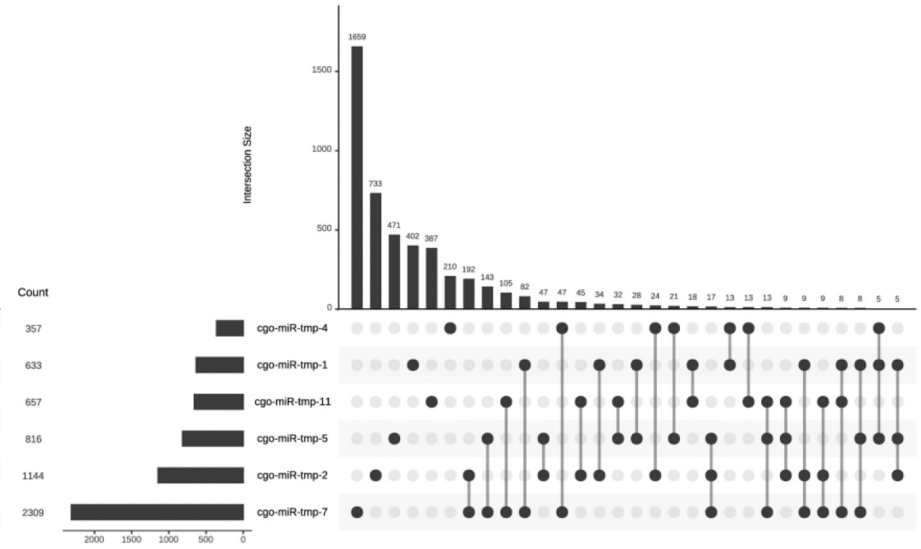


Fig. 2. Statistics of target genes of Symbiodiniaceae miRNAs in the form of Upset plot. Each column corresponds to a miRNA, and bar charts on top show the target gene number of the miRNA. Each row corresponds to a possible intersection: the filled-in cells show annotation from which miRNA is part of an intersection sharing common target genes.

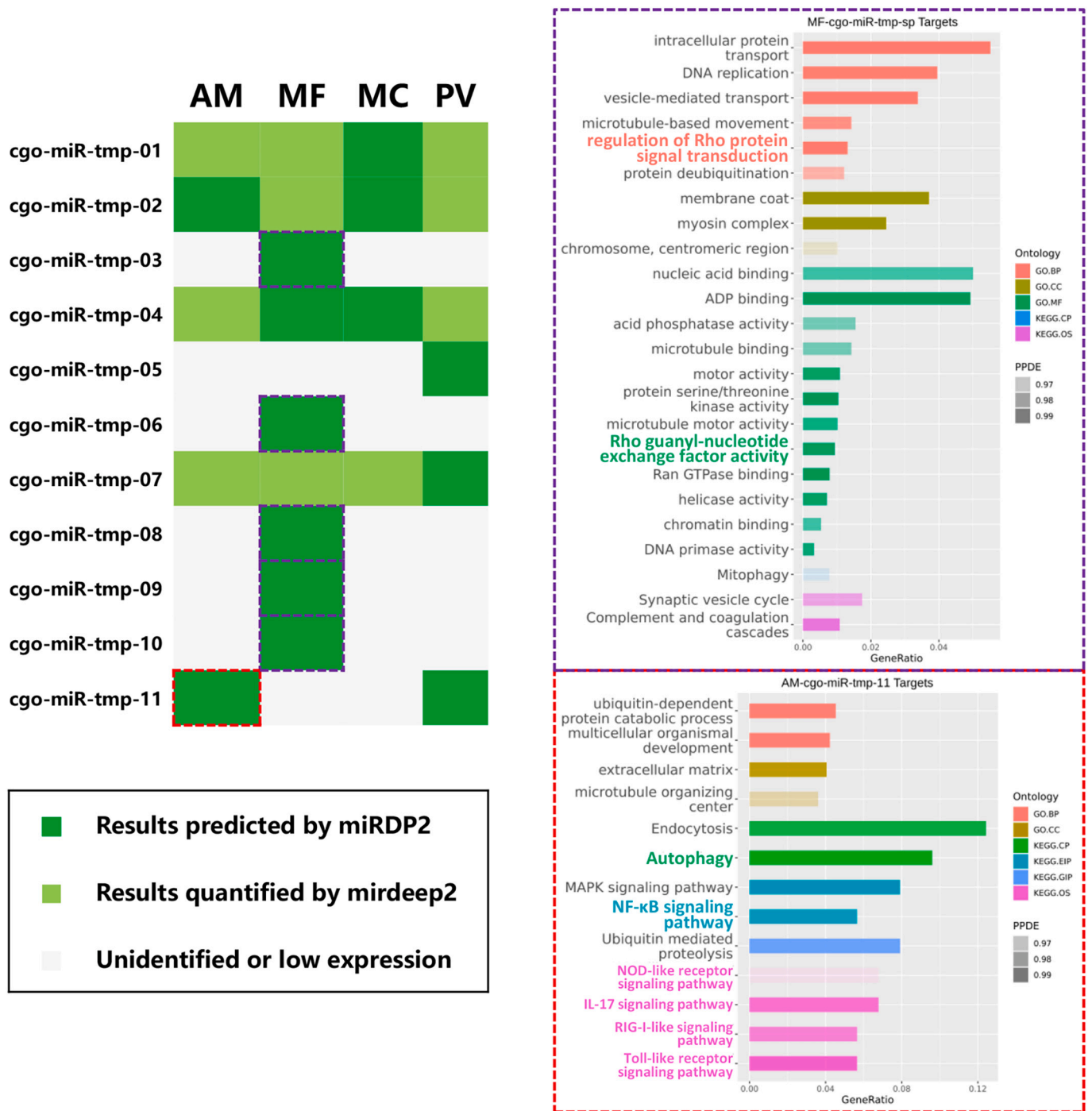


Fig. 3. Results of miRNA-targets analysis. AM, MF, MC, PV are short for *A. muricata*, *M. foliosa*, *M. capricornis*, and *P. verrucosa*. GO.BP, GO.CC, GO.MF are short for biological process, cellular component and molecular function in gene ontology. KEGG.CP, KEGG.EIP, KEGG.GIP, KEGG.OS are short for cellular processes, environmental information processing, genetic information processing and organismal systems in KEGG. Enrichment plots are generated with categories marked by color and statistical significance marked by transparency. PPDE = 1 - Pvalue. Left: identified Symbiodiniaceae miRNAs in investigated corals. The vertical axis indicates the miRNA name and the horizontal axis is the species name. Results of miRDP2 are marked with dark green while expanded results of mirdeep2 quantifier are marked with light green. Upper right: enrichment results of cgo-miR-tmp-3,6,8,9,10 targets in *M. foliosa*. Notable terms include a series of GO terms related to cytoskeleton as well as cell migration, cell adhesion, and cell proliferation, including “regulation of Rho protein signal transduction,” “DNA replication,” and “microtubule-based movement” in biological process, “myosin complex” and “chromosome centromeric region” in cellular component and “Rho guanyl-nucleotide exchange factor activity,” “Ran GTPase binding,” “motor activity,” and “chromatin binding” in molecular function. Lower right: enrichment results of cgo-miR-tmp-11 targets in *A. muricata*. Enriched pathways are immune-related pathways including NOD-like receptor signaling pathway, IL-17 signaling pathway, RIG-I-like signaling pathway and Toll-like receptor signaling pathway. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

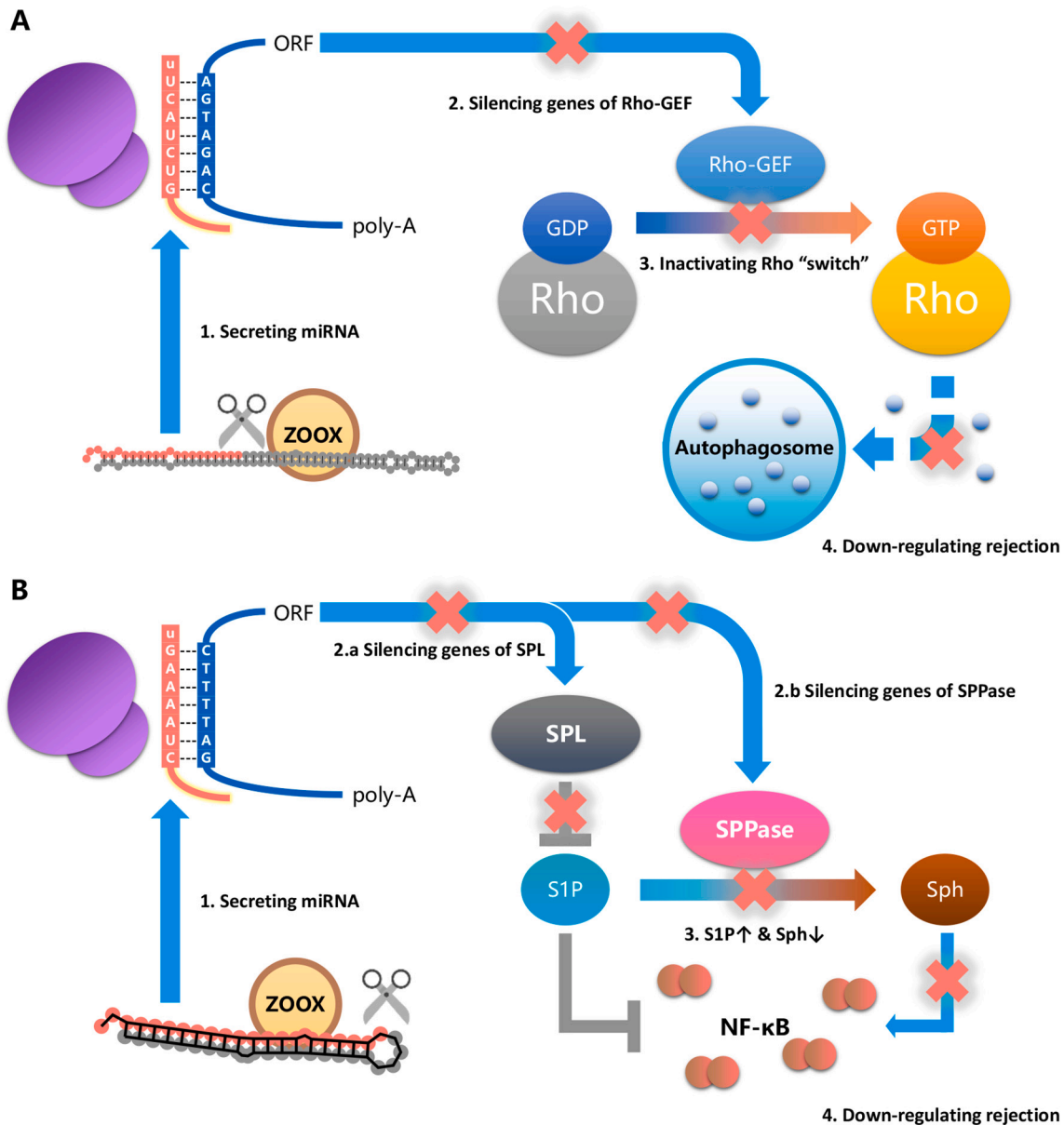


Fig. 4. Predicted downregulating rejection pathway in coral-Symbiodiniaceae holobionts. ZOOX is short for Symbiodiniaceae. **A.** Symbiodiniaceae miRNAs target and silence coral Rho-GEF genes to prevent the activation of Rho GTPases to block threatening autophagosome production in the coral. **B.** Symbiodiniaceae miRNAs inhibit the expression of SPL for S1P degradation and SPPase for Sph production, so as to downregulate NF- κ B related to apoptosis.

Committee of Institutional Animal Care and Use Committee of Nanjing Medical University (protocol code IACUC-1910003 and date of approval is 10 October 2019).

The species in the study were collected from the Xisha Islands in the South China Sea (latitude 15°40′–17°10′ north, longitude 111°–113° east).

The coral samples were cultured in our laboratory coral tank with conditions conforming to their habitat environment. All the species were raised in a RedSea® tank (redsea575, Red Sea Aquatics Ltd) at 26 °C and 1.025 salinity (Red Sea Aquatics Ltd). The physical conditions of the coral culture system are as follows: three coral lamps (AI®, Red Sea Aquatics Ltd), a protein skimmer (regal250s, Reef Octopus), a water chiller (tk1000, TECO Ltd), two wave devices (VorTech™ MP40, EcoTech Marine Ltd), and a calcium reactor (Calreact 200, Reef Octopus), etc.

5.2. Method details

5.2.1. RNA extraction

All RNA extraction procedures follow the manufacturers' instructions. The total RNA was isolated with TRIzol LS Reagent (Thermo Fisher Scientific, 10,296,028) and treated with DNase I (Thermo Fisher Scientific, 18,068,015). The high-quality mRNA was isolated with a FastTrack MAG Maxi mRNA Isolation Kit (Thermo Fisher Scientific, K1580-02). Samples were separated from healthy *Acropora muricata*, *Montipora foliosa*, *Montipora capricornis* and *Pocillopora verrucosa* to ensure that enough high-quality RNA—more than 10 μ g—could be obtained for a cDNA transcriptome library.

5.2.2. Full-length transcriptome sequencing

Before establishing a cDNA transcriptome library, the quality of total RNA had to be tested. Agarose gel electrophoresis was used to analyze the degree of degradation of RNA and whether it was contaminated. A

Table 2

Target genes related to S1P and Sph. Details of genes and targeting sites are available in **File S4** as well as online database CoralSym (<http://coral.bmeonline.cn/sym/target/>).

Species	Gene ID	Annotation	miRNA	Target ID
<i>A. muricata</i>	4965	Sphingosine-1-phosphate phosphatase 2-like	cgo-miR-tmp-7	19294
			cgo-miR-tmp-8	2879
	23507	Sphingosine-1-phosphate lyase 1-like	cgo-miR-tmp-7	26812
<i>M. foliosa</i>	23525	Sphingosine-1-phosphate lyase 1-like	cgo-miR-tmp-7	26832
	23421	Sphingosine-1-phosphate lyase 1-like	cgo-miR-tmp-7	25133
	23465	Sphingosine-1-phosphate lyase 1	cgo-miR-tmp-7	25158
	13430	Sphingosine-1-phosphate lyase 1-like	cgo-miR-tmp-7	32740
<i>M. capricornis</i>	13441	Sphingosine-1-phosphate lyase 1-like	cgo-miR-tmp-7	32768
	13471	Sphingosine-1-phosphate lyase 1-like	cgo-miR-tmp-7	32829
	13478	Sphingosine-1-phosphate lyase 1-like	cgo-miR-tmp-7	32849
	7577	Sphingosine-1-phosphate phosphatase 2	cgo-miR-tmp-7	9979

Nanodrop nucleic acid quantifier was used to detect the purity of RNA (OD260/280 ratio), a Qubit RNA assay was used to quantify the RNA concentration accurately, and an Agilent 2200 TapeStation was used to accurately detect the integrity of the RNA.

The Clontech SMARTer® PCR cDNA Synthesis Kit (Clontech Laboratories, 634,926) and the BluePippin Size Selection System protocol, as described by Pacific Biosciences (PN 100–092–800-03) were using to prepare the Iso-Seq library according to the Isoform Sequencing protocol (Iso-Seq).

The RNA-seq library was prepared according to the standard library construction for the Illumina HiSeq X Ten, using three biological replicates for each species of coral.

We used the PacBio Sequel II platform with SMRT (single molecular real time) sequencing technology and SMRTlink v7.0 software (minLength 50; maxLength 15,000; minPasses 1) to process sequencing samples [42]. After polymer read bases were performed, the subreads. bam files were obtained by removing the joint and the original offline data with lengths of under 50 bp. The CCSs (circular consensus sequences) were obtained by putting the subreads.bam file through the CCS algorithm, which is self-correcting for the single molecule multiple sequencing sequence. Consequently, the full-length-non-chimera (FLNC) and nFL (non-full-length, non-chimera) sequences were identified by detecting whether CCSs contained 5'-primer, 3'-primer and poly-A. FLNC sequences of the same transcript were clustered with a hierarchical $n * \log(n)$ algorithm to obtain consensus sequences. The corrected consensus reads were polished from consensus sequences (Arrow polishing) using LorDEC [43] v0.7 software using the RNA-seq data sequenced with the Illumina HiSeq X Ten platform. Using CD-HIT [44] software ($-c 0.95 -T 6 -G 0 -aL 0.00 -aS 0.99$), all redundancies were removed in corrected consensus reads to acquire final full-length transcripts and unigenes for subsequent bioinformatics analyses.

5.2.3. Small RNA sequencing

A total amount of 3 µg of RNA per sample was used as input material for the small RNA library. Sequencing libraries were generated using the NEBNext® Multiplex Small RNA Library Prep Set for Illumina® (NEB, USA) following the manufacturer's recommendations, and index codes were added to attribute sequences to each sample. Briefly, the NEB 3' SR Adapter was directly and specifically ligated to the 3' end of miRNA,

siRNA and piRNA. After the 3' ligation reaction, the SR RT Primer was hybridized to the excess of 3' SR Adapter that remained free after the 3' ligation reaction, transforming the single-stranded DNA adapter into a double-stranded DNA molecule. This step is important to prevent adapter-dimer formation, and dsDNAs are also not substrates for ligation mediated by T4 RNA Ligase 1, so they do not ligate to the 5' SR Adapter in the subsequent ligation step. Next, the 5' ends adapter was ligated to the 5' ends of miRNAs, siRNA and piRNA. Then the first strand of cDNA was synthesized using M-MuLV Reverse Transcriptase (RNase H-). PCR amplification was performed using LongAmp Taq 2× Master Mix, SR Primer for Illumina and index (X) primer. PCR products were purified on an 8% polyacrylamide gel (100 V, 80 min). DNA fragments corresponding to 140–160 bp (the length of small noncoding RNA plus the 3' and 5' adapters) were recovered and dissolved in 8 µL elution buffer. Finally, library quality was assessed with the Agilent Bioanalyzer 2100 system using DNA High Sensitivity Chips.

The clustering of the index-coded samples was performed on a cBot Cluster Generation System using a TruSeq SR Cluster Kit v3-cBot-HS (Illumina) according to the manufacturer's instructions. After cluster generation, the library preparations were sequenced on an Illumina HiSeq X Ten platform and 50 bp single-end reads were generated.

Clean data for downstream analysis were obtained by removing reads with low quality, poly-N|A|T|G|C, and 5' adapter contaminants. Reads without the 3' adapter or the insert tag were cleaned as well.

5.2.4. Gene function annotation

Gene function was annotated using the following databases: Nr (NCBI non-redundant protein sequences) [45], Nt (NCBI non-redundant nucleotide sequences), Pfam (Protein family) [46], KOG (Clusters of Orthologous Groups of proteins) [47], Swiss-Prot (A manually annotated and reviewed protein sequence database) [48], GO (Gene Ontology) [49] and KEGG (Kyoto Encyclopedia of Genes and Genomes) [17]. We use BLAST 2.7.1+ in NCBI setting the e-value '1e-5' in Nt database analysis, Diamond v0.8.36 BLASTX software, setting the e-value to '1e-5' in Nr, KOG, Swiss-Prot and KEGG databases analysis, and HMMER 3.1 package for Pfam database analysis. Furthermore, published Symbiodiniaceae gene models were employed to correct and supplement Nr annotation by the aid of BLASTP (2.10.0+), where hits with higher scores were identified as final annotation.

5.2.5. miRNA identification

miRDeep-P2 [50] was employed to identify Symbiodiniaceae miRNAs in each coral separately after building the bowtie index of the *Cladocypium goreau* genome [22] downloaded from Reefgenomics [18]. Initial outputs were corrected by genome annotation in which sequences with loci hitting CDS were removed. Then miRDeep2 [51] using filtered sequences as inputs was run on all raw data to quantify miRNA expression as well to eliminate false negatives. Finally, their locations on another two *Cladocypium* genomes were searched by the aid of BLASTN (2.10.0+).

5.2.6. Enrichment analysis

All miRNA targeting sites were predicted with miRanda [52] ($-sc 145 -en -10 -scale 4 -strict$). Predicted target genes were all selected for GO/KEGG enrichment using clusterProfiler [53] ($pvalueCutoff = 0.05$, $pAdjustMethod = 'BH'$, $qvalueCutoff = 0.2$).

5.2.7. Website development

CoralBioinfo (<http://coral.bmeonline.cn>) was developed under the LEMP (Linux, Nginx, MySQL, and PHP) stack. Web functions were implemented with PHP 7.2, while Nginx works as the web server. Most services have been built into Docker images, which are beneficial for cross-platform deployment and extended development.

All data are stored in MySQL tables with indexes to improve query efficiency. In addition, sequence data have been formatted into NCBI-BLAST local databases for alignment and download services.

Additional resources

Detailed database structure manuals along with user tutorials are all available at <http://coral.bmeonline.cn/help.php>.

API documents are available at <http://coral.bmeonline.cn/api/>.

Ethics approval and consent to participate

All coral samples were collected and processed in accordance with local laws for invertebrate protection and approved by the Ethics Committee of Institutional Animal Care and Use Committee of Nanjing Medical University (protocol code IACUC-1910003 and date of approval is 10 October 2019).

Consent for publication

Not applicable.

Availability of data and materials

Raw data from our own full-length transcriptome sequencing and small RNA sequencing are available from NCBI (PRJNA544778).

Though we sincerely recommend acquiring data via our CoralBioinfo database (<http://coral.bmeonline.cn>) introduced in this paper, we have already uploaded necessary supplementary data to better serve researchers.

Funding

This work was supported by the open research fund of State Key Laboratory of Bioelectronics, Southeast University [Sk1b2021-k02], and the open research fund program of Guangxi Key Lab of Mangrove Conservation and Utilization [GKLMC-202002].

Authors' contributions

YZ: experiment, database, writing and editing. TH: data uploading. ZL & JC: reviewing. CH: supervision. XL: project approval. All authors contributed to the article and approved the submitted version.

Declaration of Competing Interest

The authors declare that they have no competing interests.

Acknowledgements

We are grateful for Dr. Xiaojun Xia's technical support in our production environment.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ygeno.2022.110404>.

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